National Postsecondary Certificate Trends

Research Progress Report | Week 4

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1 Introduction

Certificates are the smallest credentials that colleges hand out. They can be as short as a few weeks or as long as two years. Most people think of college as a place that gives out degrees, but a big share of what US colleges actually produce every year is certificates. In 2024, about one in every four credentials awarded in the country was a certificate, not a degree.

This project looks at what has been happening with certificates in the United States over the last twenty five years? Who is producing them, in what fields, and how has the mix changed? To answer this, I built a national dataset covering every Title IV institution from 2000 to 2024, using the IPEDS Completions files. The dataset tracks awards by institution, field of study, and credential type. It includes both certificates and traditional degrees, so we can compare how they moved together or apart over time.

Before asking why something is happening, we want a clear picture of what is happening. This report walks through that picture. It covers the growth of certificates compared to degrees, the mix of certificate types, the fields that dominate certificate production, and the kinds of institutions doing most of the work. There are a few patters that are worth noting. Certificates grew faster than degrees over the period, and the gap widened after 2020. The for-profit sector, which was a large certificate producer in the late 2000s, shrank sharply after 2011 and has only partially recovered. Public institutions have become the dominant certificate producer by a wide margin. Health remains the single largest field, but its share of certificates is falling.

This report describes the data and how the panel was built, show the national trends, and looks at the mix of certificate types and the fields that drive certificate production. It also breaks the story down by institution type and discusses some limitations and open questions.

2 Data and Panel Construction

2.1 Source

The data come from the Integrated Postsecondary Education Data System (IPEDS), which is the federal government's main source of information on colleges and universities. Every institution in the United States that participates in federal student aid programs is required to report to IPEDS every year. This means the panel covers essentially every college that matters for a national story about credentials.

I used two IPEDS files for each year from 2000 to 2024. The first is the Completions file (C), which reports the number of awards each institution gave out, broken down by field of study (using CIP codes) and by award level. The second is the Institutional Characteristics file (HD), which provides basic information about each institution, including whether it is public, private nonprofit, or for-profit, and whether it is a four-year, two-year, or less-than-two-year school. Merging the two gives us awards at the institution level tagged with the institution's type.

2.2 Panel structure

The final dataset has one row for every unique combination of institution, two-digit field of study, award level, and year. Each row records the total number of first-major completions in that cell. The panel covers 25 years, roughly 6,000 to 7,000 institutions per year, and 42 two-digit CIP fields. Second majors are excluded because counting them would double-count students who complete a double major, which is not what we want when tracking credential production.

The panel is unbalanced. Schools open, close, and merge over the period, and the CIP classification system was revised in 2000, 2010, and 2020. Both of these introduce discontinuities that are noted and discussed further here.

2.3 Award categories

The IPEDS award level codes changed over time. Before 2011, the "less than one year" certificate was a single bucket (code 1). In 2011, IPEDS split this into two finer categories: certificates of less than 12 weeks (code 20) and certificates of 12 weeks to less than one year (code 21). Data on the split, however, is only reliably available starting in 2020. Doctorate coding also changed. Before 2008, doctorates were reported as a single category (code 9), along with first-professional degrees (codes 10 and 11). Starting in 2008, IPEDS introduced three doctorate types (codes 17, 18, and 19), and the first-professional category was folded into the professional practice doctorate (code 18).

To make the categories comparable across the full 25 years, I built a harmonized crosswalk. Old code 1 was merged with codes 20 and 21 into a single "under one year" bucket. Old doctorate codes 9, 10, and 11 were merged with new codes 17, 18, and 19 into a single doctorate bucket. The full set of categories used in this report is: five certificate types (under one year, one to two years, two to four years, postbaccalaureate, and post-master's) and four degree types (associate, bachelor's, master's, and doctorate).

2.4 Institution categories

Institutions are classified along two dimensions. CONTROL identifies whether a school is public, private nonprofit, or private for-profit. ICLEVEL identifies whether the school offers four-year programs, two-year programs, or less-than-two-year programs. Combining the two gives a nine-cell grid that covers the main institutional types in US higher education, from public community colleges to for-profit trade schools to research universities.

2.5 Benchmarking

Building a panel from raw data over 25 years introduces many chances for something to go wrong. Field naming conventions change. Variable types shift between numeric and string. Summary rows can be double-counted. To catch these issues, I benchmarked the panel every year against the totals published by the National Center for Education Statistics in the Digest of Education Statistics. Bachelor's degree totals match published NCES figures within about two percent for every year from 2000 to 2024. Associate, master's, and doctorate totals also align closely with published sources. This gives me reasonable confidence that the panel captures the full national picture.

Availability

All code and cleaned output files are available at https://github.com/fedamohammadi/certificate_trends. Raw IPEDS files are not included in the repository because they are large and are already publicly available on the NCES website. A short README in the repository lists which files to download to reproduce the pipeline from scratch. The final panel is titled `completions_panel_2000_2024.dta` and is in the `data/clean/` folder. The five tables that support this report (referenced throughout Section 3 and Section 4) are in the `output/tables/` folder in CSV format.

3 National Trends in Certificates and Degrees

3.1 The big picture

Between 2000 and 2024, total credentials awarded by US colleges more than doubled. In 2000, colleges handed out about 3.1 million awards. By 2024, that number was 5.5 million. Certificates and degrees both grew, but they did not grow at the same rate or at the same time. Figure 1 tracks how each has changed since 2000, with both series set to 100 in that year.

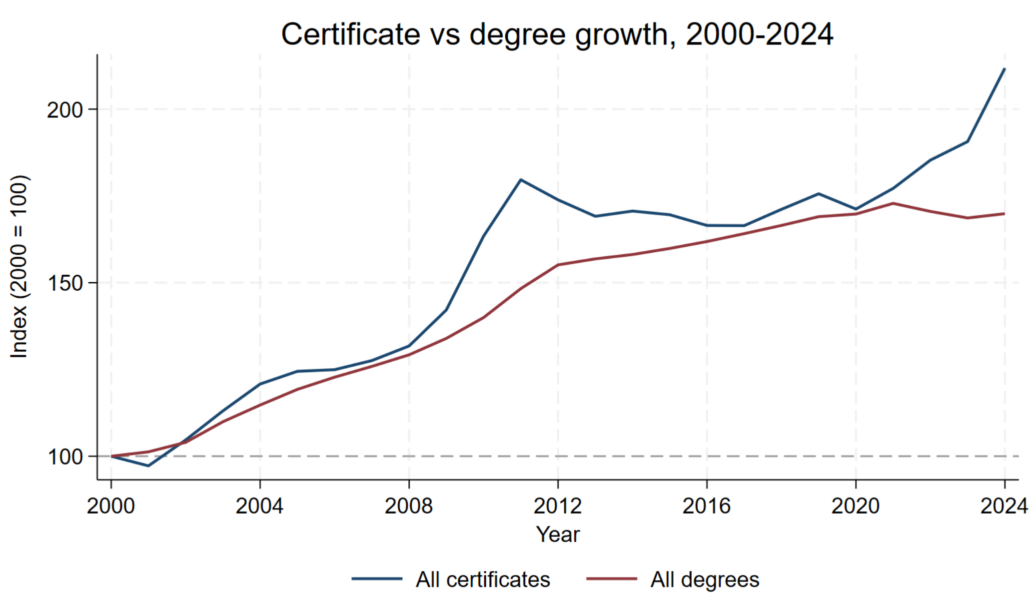

Figure 1: Certificate awards grew faster than degrees over the full period, with a widening gap after 2020. Both series are indexed to 2000 = 100.

Degrees grew steadily. By 2024, colleges were awarding about 70 percent more degrees than they did in 2000. Certificates grew faster. By 2024, certificate production was more than twice the 2000 level. The gap between the two lines is what matters here. For most of the 2000s, certificates and degrees tracked each other. Starting around 2009, certificates pulled ahead, driven by a sharp increase during the recession years. They gave back some of that gain in the mid-2010s, then began climbing again after 2020. By 2024, the gap between certificates and degrees was the widest it had been in the entire period.

3.2 Certificate share of all awards

If certificates are growing faster than degrees, we would expect certificates to make up a bigger share of all awards over time. Figure 2 shows that this happens, but only slightly and only recently.

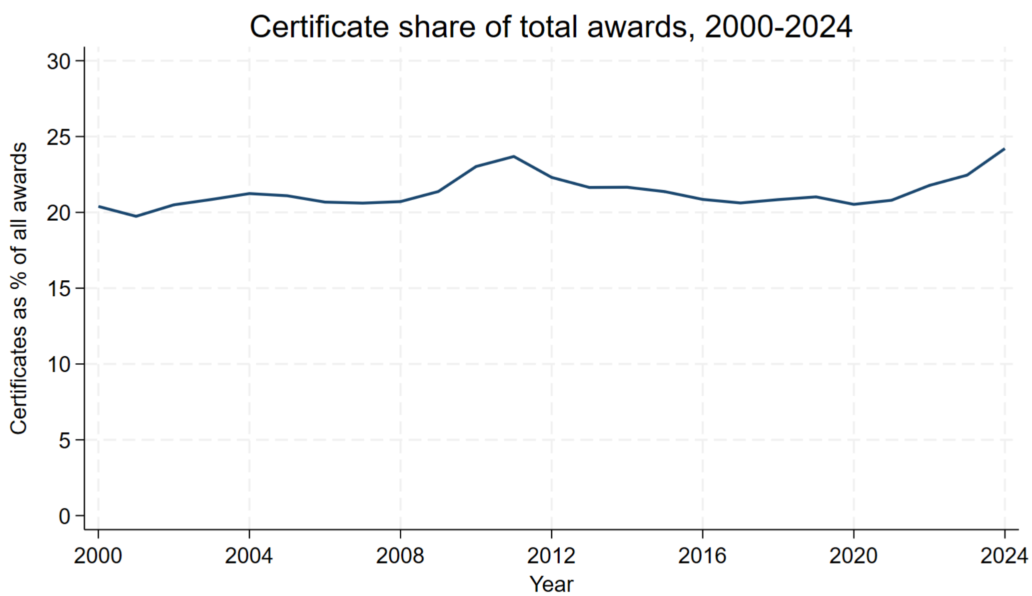

Figure 2: Certificates make up about one in five awards for most of the period, with a small bump around 2011 and a clearer rise starting in 2022.

For most of the last 25 years, certificates made up between 20 and 22 percent of all awards. There was a small spike around 2010 and 2011 during the recession, when certificate enrollment surged. Then the share settled back down and stayed near 21 percent through the rest of the 2010s. Starting in 2022, the share began climbing again, reaching 24 percent by 2024. This is the highest certificate share of the entire period.

This is a slower and more modest shift than Figure 1 might suggest. Both certificates and degrees have been growing, so the ratio between them changes slowly even when their growth rates differ. The clear pull-away in the growth chart shows up as only a two or three percentage point rise in the share chart. Both are true, and both matter, but they answer slightly different questions.

3.3 Mix of certificate types

Zooming in on certificates alone, the five types have very different sizes and trajectories. Figure 3 shows all five over the full period.

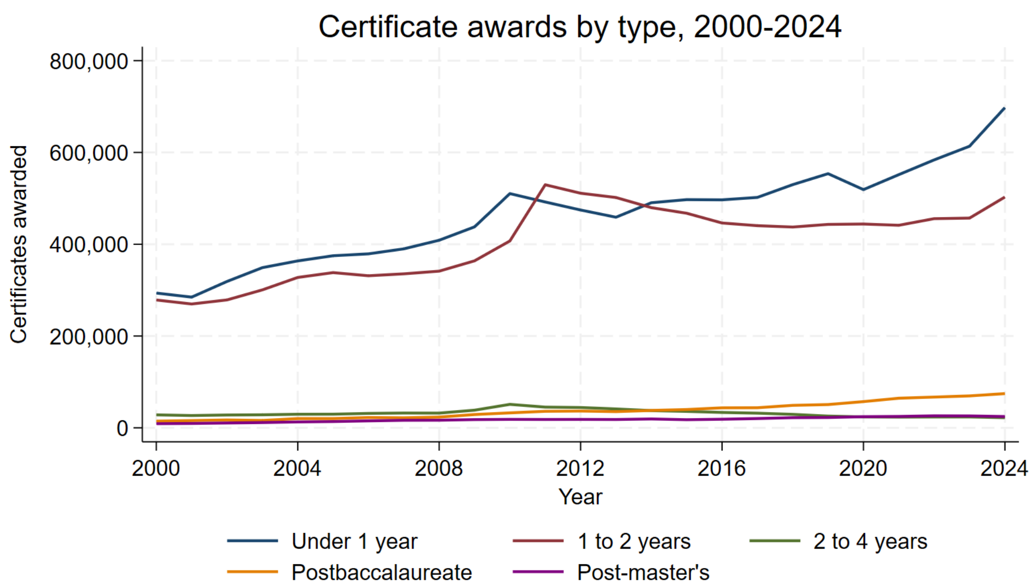

Figure 3: Under-one-year certificates and one-to-two-year certificates dominate. The two-to-four-year, postbaccalaureate, and post-master's categories are much smaller but the postbaccalaureate category grew fivefold.

Two categories dominate. Under-one-year certificates grew from about 294,000 in 2000 to 698,000 in 2024. One-to-two-year certificates grew from 279,000 to 503,000 over the same period. Together, these two categories account for the vast majority of all certificate production in every year of the panel.

The other three categories are much smaller in absolute terms, but they tell interesting stories of their own. The two-to-four-year certificate category has been essentially flat, moving from 28,000 in 2000 to 22,000 in 2024. Post-master's certificates grew modestly, from 9,000 to 25,000. The postbaccalaureate certificate category is the standout. It grew from 15,000 in 2000 to 75,000 in 2024, a fivefold increase. This is the fastest growth rate of any credential category in the panel, certificate or degree. It suggests that colleges are increasingly offering short credentials to people who already have bachelor's degrees, likely as a way to add a specific skill or credential to an existing career path.

The under-one-year line also has a distinctive shape. It grew steadily through the 2000s, jumped sharply during the recession, plateaued in the mid-2010s, and then took off again after 2020. The post-2020 acceleration is worth watching because it is happening at a speed that outpaces any earlier period in the panel.

4 Who Is Producing Certificates, and In What Fields

4.1 Public institutions dominate, for-profits have a distinct arc

The overall growth in certificates hides a lot of variation across institution types. Figure 4 breaks certificate production down by whether the institution is public, private nonprofit, or private for-profit.

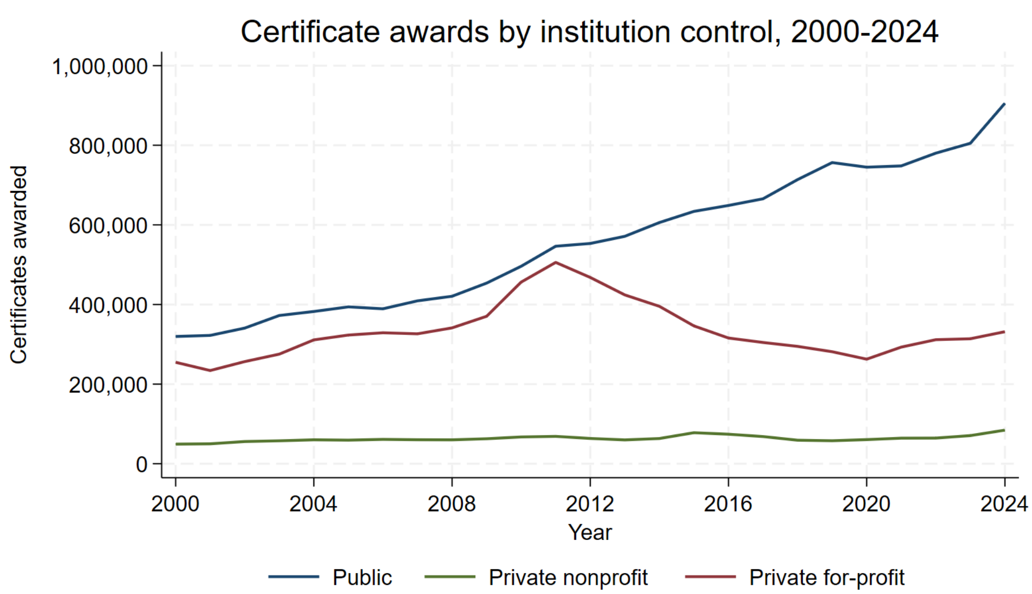

Figure 4: Public institutions have become the dominant certificate producer. For-profits peaked around 2011 and then collapsed, with a partial recovery beginning in 2020.

Public institutions have led certificate production for the entire period and their lead has grown. In 2000, public colleges awarded about 320,000 certificates. By 2024, that number was 900,000, nearly tripling over the 25 years. This growth has been steady and continuous, with a small dip during the pandemic and a strong recovery afterward.

The for-profit story is very different. For-profit institutions were awarding about 250,000 certificates in 2000. Their output grew rapidly through the 2000s and peaked around 500,000 in 2011. Then it collapsed. By 2020, for-profit certificate production had fallen to about 265,000, roughly half of the peak level. The collapse lined up with a period of tighter federal regulation of the for-profit sector, including the gainful employment rule and the shutdowns of several large chains such as Corinthian Colleges and ITT Technical Institute. Starting around 2020, for-profits began to recover, and by 2024 they were awarding about 330,000 certificates. This is still well below the 2011 peak.

Private nonprofit institutions have been a minor player in certificate production throughout the period. Their output has held roughly steady between 50,000 and 80,000 per year and has never approached the volume of either public or for-profit schools. The certificate story in the US is mostly a story about public colleges and for-profits.

More detail on the split by institution level (four-year, two-year, less-than-two-year) is available in Table 3 in the repository, which shows the full 3-by-3 grid for four selected years: 2000, 2010, 2019, and 2024.

4.2 Short certificates within the under-one-year category

Starting in 2020, IPEDS began separately reporting certificates that take less than 12 weeks and certificates that take 12 weeks to less than one year. Before 2020, both were reported as a single "under one year" bucket. Figure 5 shows the split for the five years we have data.

The 12-weeks-to-1-year category is by far the larger of the two. In 2020, it accounted for 432,000 awards. By 2024, it had grown to 582,000. The under-12-weeks category is smaller but not trivial, with about 88,000 awards in 2020 growing to 116,000 in 2024. Both categories are climbing. The 12-week-plus category grew by about 35 percent over five years, and the sub-12-week category grew

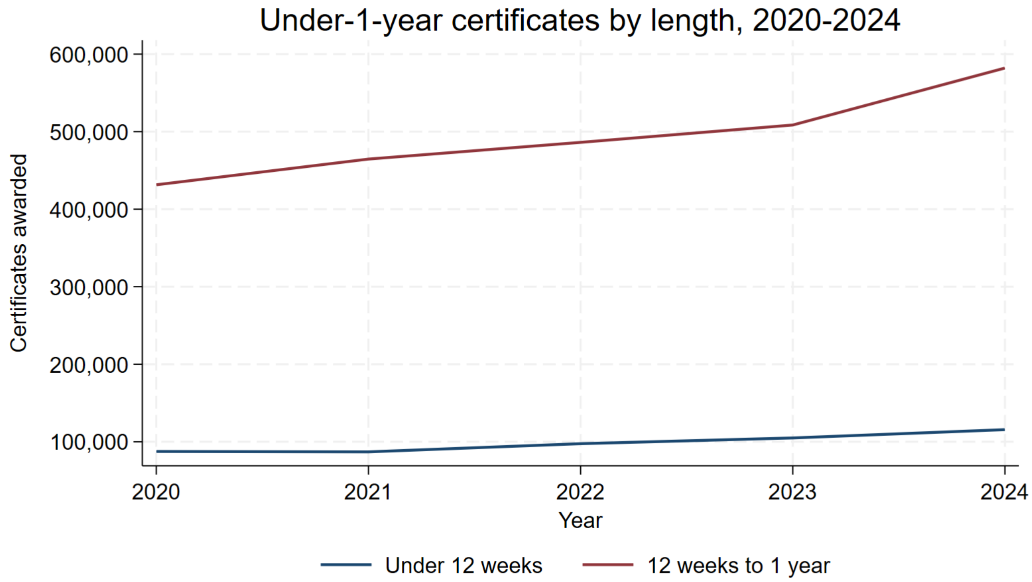

Figure 5: The 12-weeks-to-1-year category is about five times larger than the under-12-weeks category. Both are growing, and the 12-weeks-to-1-year category is growing faster.

by about 32 percent.

Five years is not enough time to identify a stable trend, but the direction is consistent with what we see in the broader post-2020 certificate boom shown in Figure 3. Short-form credentials are gaining ground quickly.

4.3 Fields of study for undergraduate certificates

Certificates are concentrated in a small number of fields. Table 4 in the repository lists the top 15 two-digit CIP fields for undergraduate certificates in 2015, 2019, and 2024. A few patterns are worth highlighting.

Health professions (CIP 51) is the single largest field in every year of the panel, but its share of undergraduate certificates has been falling. In 2015, health accounted for about 35 percent of all undergraduate certificates. By 2024, its share had dropped to 24 percent. This is not because health certificate production shrank in absolute terms. It stayed roughly flat around 290,000 to 350,000 per year. Health's share fell because other fields grew faster.

Personal and culinary services (CIP 12) has held steady at around 12 to 14 percent, driven mostly by cosmetology programs. Business (CIP 52) has been around 10 percent. Mechanic and repair technologies (CIP 47) has held about 8 percent. These are the traditional workforce-oriented certificate fields, and they have been stable over the last decade.

The field that has grown fastest in the certificate mix is liberal arts and general studies (CIP 24). It went from 4.5 percent of undergraduate certificates in 2015 to 11.2 percent in 2024. In absolute terms, certificates in this category grew from about 45,000 to 137,000. This is probably a real trend in how community colleges are packaging transfer-oriented credentials rather than a coding artifact, since the underlying category (across all award levels) has been growing steadily since 2000.

Computer and information sciences (CIP 11) also grew as a share of certificates, going from 3.5 percent to 5.4 percent between 2015 and 2024. Construction trades (CIP 46) grew from 2.5 percent to 3.9 percent. Both of these gains come from real absolute growth, not just a shift in the mix.

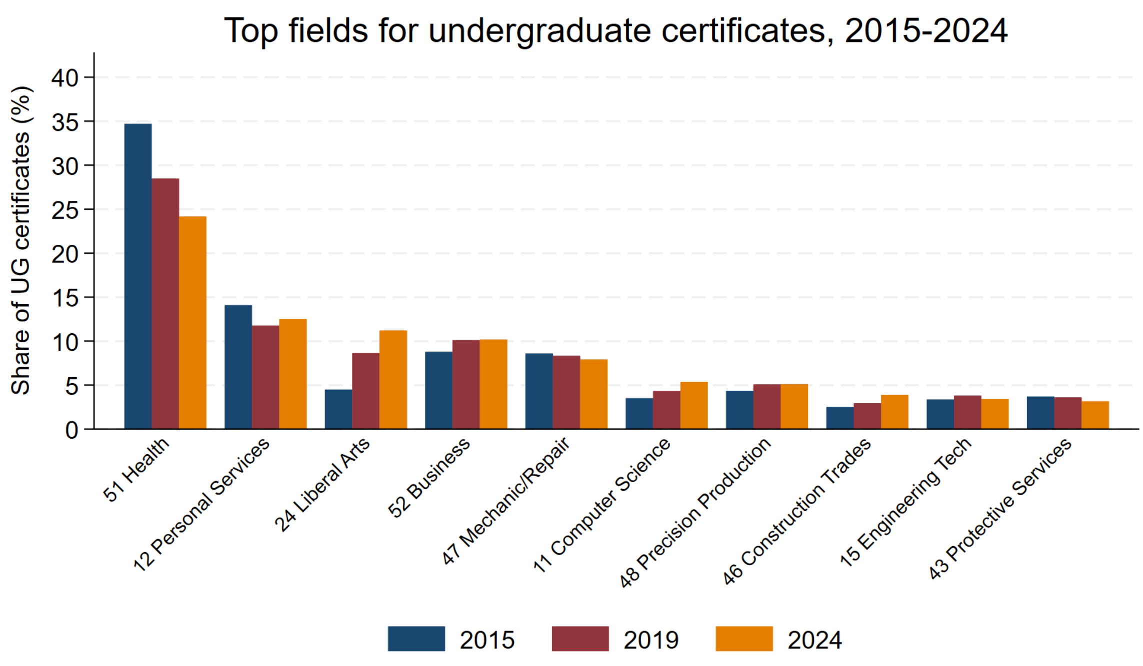

Figure 6: Share of undergraduate certificates by two-digit CIP field, 2015, 2019, and 2024. Fields are sorted by 2024 share. Only fields with at least a 3 percent share in one of the three years are shown.

4.4 A note on the 2-digit CIP level

The two-digit CIP classification works well for most of the fields discussed above. Fields like construction trades, mechanic and repair, or business are internally coherent enough that the two-digit view captures what is happening. Health is a notable exception. CIP 51 includes everything from short phlebotomy programs and medical assisting certificates to two-year nursing degrees. A four-digit breakdown within CIP 51 would give a much clearer picture of which specific health credentials are driving the totals. Personal and culinary services (CIP 12) has a similar issue, since it mixes cosmetology, culinary arts, and funeral services. These are worth flagging as areas where a finer-grained look would be valuable in later analysis.

Beyond individual fields, the CIP classification system was revised in 2000, 2010, and 2020. Most two-digit codes are stable across these revisions, but a handful shift. Table 5 in the repository documents which codes are present in which years, so you can see where there are potential discontinuities. For the fields highlighted in this section, the two-digit codes are stable across the full period.

5 Limitations and Open Questions

Everything in this report is descriptive. The figures show what has happened to certificate production over 25 years, not why. The for-profit collapse after 2011 lines up with the timing of the gainful employment rule and the Corinthian and ITT closures, but this report does not establish causation. The fivefold rise in postbaccalaureate certificates is consistent with rising demand for short credentials from degree holders, but I did not test that against alternatives. Causal analysis is possible using the panel, but it needs a research design beyond the scope of this first pass.

5.1 Definitional and measurement issues

Three issues are worth mentioning.

First, the under-one-year certificate category mixes two very different products, and IPEDS only lets us tell them apart starting in 2020. Before 2020, a 6-week medical assisting certificate and an 11-month cosmetology certificate were filed under the same code. Any pre-2020 story about "short certificates" is really a story about a heterogeneous category.

Second, on the two-digit CIP classification. Most fields work well at this level, but Health (CIP 51) covers everything from phlebotomy to registered nursing, and Personal and Culinary Services (CIP 12) mixes cosmetology, culinary arts, and funeral services. Any follow-up work on health or workforce credentials should start with a finer CIP breakdown.

Third, the CIP system was revised in 2000, 2010, and 2020. Most two-digit codes are stable across these revisions, but a handful shift. Table 5 in the repository documents which codes appear in which years. Deeper analysis using less common CIP codes should check the crosswalk carefully.

5.2 What the panel could be used for next

The panel was built for descriptive analysis but is structured to support follow-on questions.

The for-profit collapse and partial recovery is the clearest natural experiment in the data. The regulatory timeline is well-documented, the affected institutions are identifiable, and public and nonprofit schools provide a comparison group. A difference-in-differences or synthetic control design could estimate how much of the decline was driven by regulation.

The post-2020 acceleration in short and postbaccalaureate certificates raises a different question. Something changed in the certificate market around 2020, and the change was not uniform across credential types. Whether this is a lasting shift or a post-pandemic blip will take more years of data, but the panel is positioned to track it.

Finally, merging the panel with county or metro-level labor market data would allow a look at whether certificate production responds to local economic conditions or moves with national trends. A four-digit CIP breakdown of health and personal services credentials would answer a question this report cannot: which specific credentials are driving the totals.

Data and Code Availability

Everything related to this project, including the raw and cleaned data files, codes, analyses, figures, and tables shown here, is available in the GitHub repository. Anyone with access can rerun everything from the raw files through the final outputs and reproduce these results.